

Structural Analysis

BRANCH: CIVIL ENGG. (CE)

COURSE CODE: CE201

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Structural Analysis (CE201) under the Civil Engg. (CE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Indeterminacy & Energy Methods

Static and Kinematic indeterminacy of beams, trusses, and frames. Strain energy due to axial load, bending, shear. Castigliano's theorems, Unit Load method for deflection.

Unit 2: Influence Lines & Moving Loads

Influence lines for reactions, shear, bending moment in simple beams. Moving loads: maximum BM, SF, focal length. ILD for trusses.

Unit 3: Arches & Cables

Three-hinged arches: horizontal thrust, radial shear, normal thrust. Two-hinged arches, temperature effect. Analysis of suspension cables, tension forces.

Unit 4: Displacement Methods

Slope-Deflection method. Moment-Distribution method. Analysis of continuous beams and portal frames. Introduction to stiffness and flexibility matrix methods.

*Castigliano's theorem: Deflection = dU/dP . Slope-Deflection: $M_{ab} = M'_{ab} + 2EI/L * (2\theta_A + \theta_B - 3\delta/L)$.*

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



